US 20250276538A1

(19) United States
(12) Patent Application Publication
Yamada
(10) Pub. No.: US 2025/0276538 A1
(43) Pub. Date: Sep. 4, 2025

(54) **PRINTER AND METHOD**

(71) Applicant: **TOSHIBA TEC KABUSHIKI KAISHA**, Tokyo (JP)

(72) Inventor: **Satoshi Yamada**, Mishima Shizuoka (JP)

(21) Appl. No.: **19/049,172**

(22) Filed: **Feb. 10, 2025**

(30) **Foreign Application Priority Data**

Mar. 1, 2024 (JP) 2024-030898

Publication Classification

(51) **Int. Cl.**

B41J 29/48	(2006.01)
B41J 2/32	(2006.01)
B41J 3/407	(2006.01)
B41J 11/00	(2006.01)
B41J 11/48	(2006.01)

(52) **U.S. Cl.**

CPC ***B41J 29/48*** (2013.01); ***B41J 2/32*** (2013.01); ***B41J 3/4075*** (2013.01); ***B41J 11/0095*** (2013.01); ***B41J 11/48*** (2013.01)

(57) **ABSTRACT**

A printer in an embodiment includes a storage unit configured to store, for each of a plurality of types of roll paper, near-end information indicating whether to notify near-end of the roll paper, a paper set unit in which the roll paper can be set, a determination unit configured to determine a type of the roll paper to be used, a setting unit configured to read near-end information relating to the roll paper, the type of which was determined, from the storage unit and set the near-end information, a near-end detection unit configured to detect the near-end of the roll paper set in the paper set unit, and a near-end processing unit configured to, only if the near-end information indicating that the near-end is notified was set by the setting unit, according to detection of near-end of the roll paper by the near-end detection unit, notify the near-end.

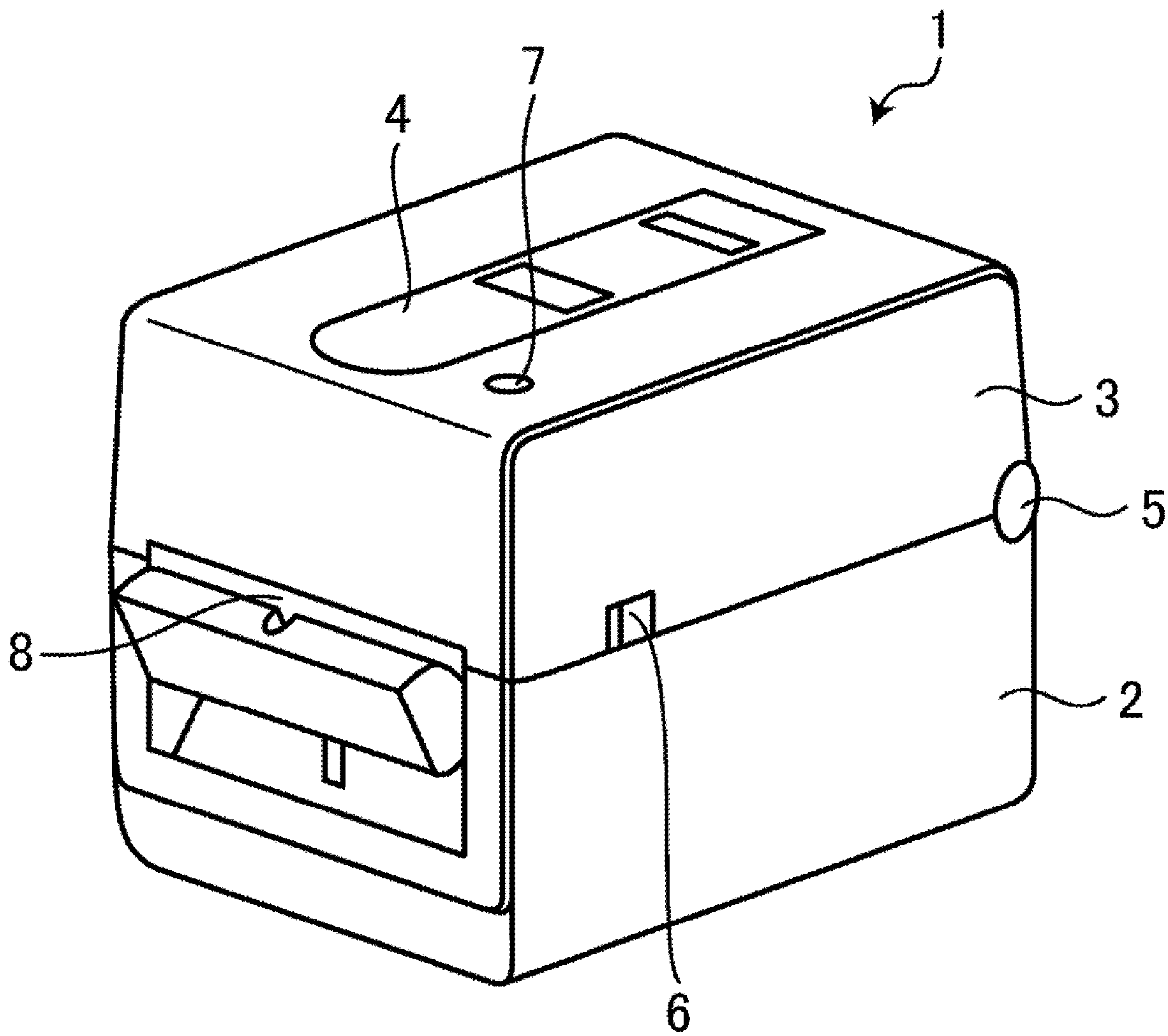


FIG. 1

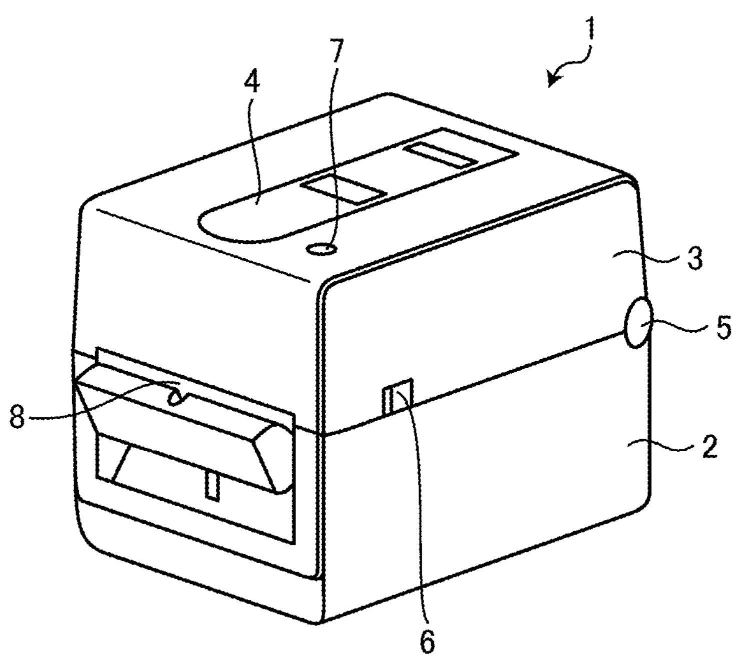


FIG. 2

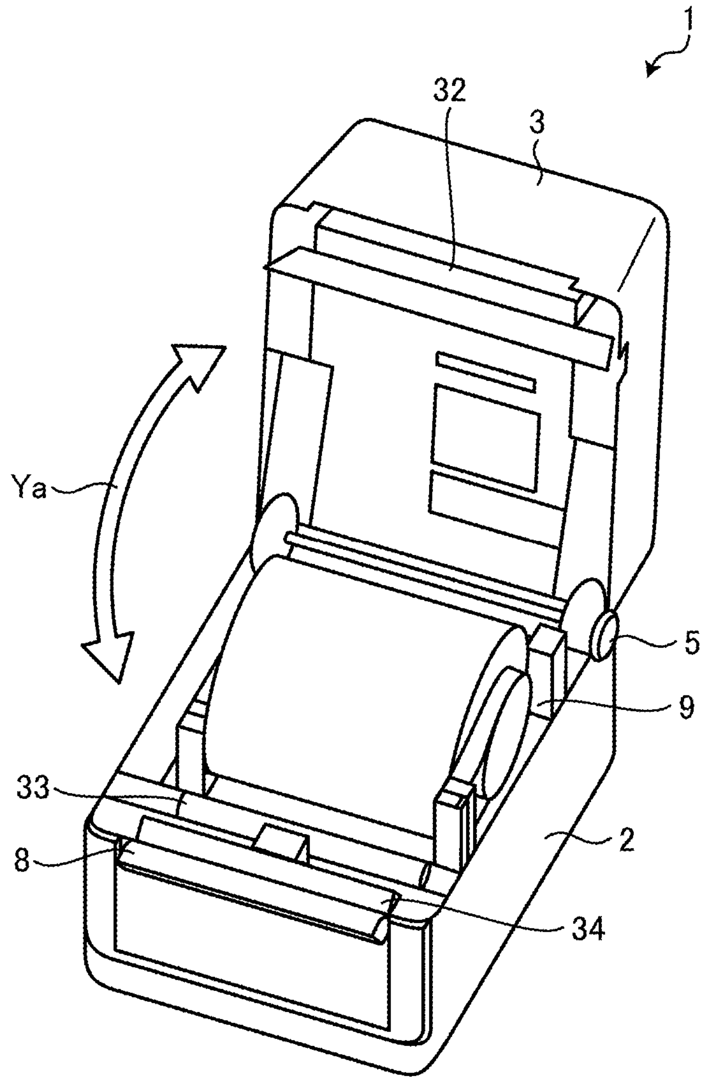


FIG. 3

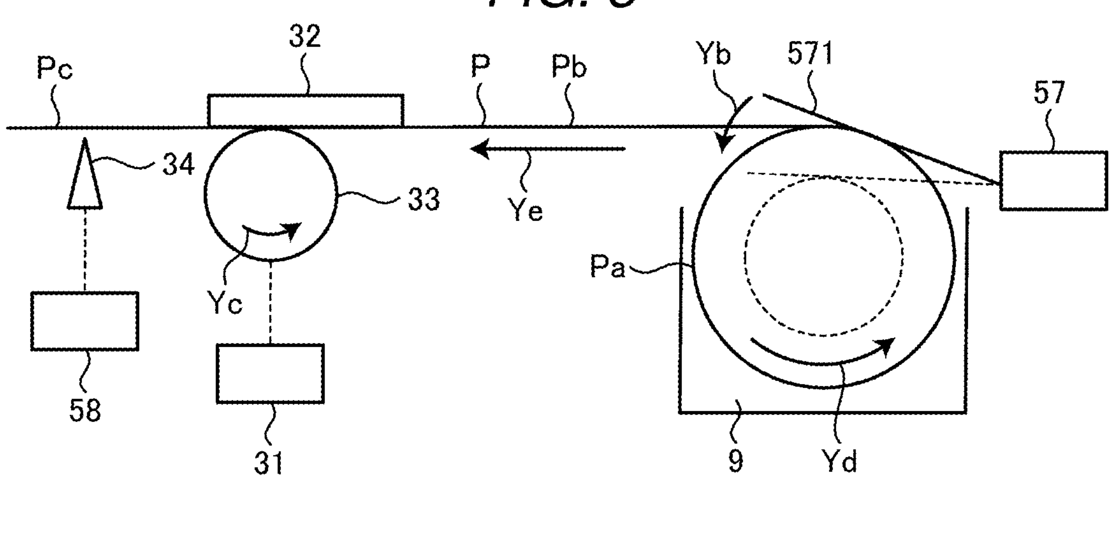


FIG. 4

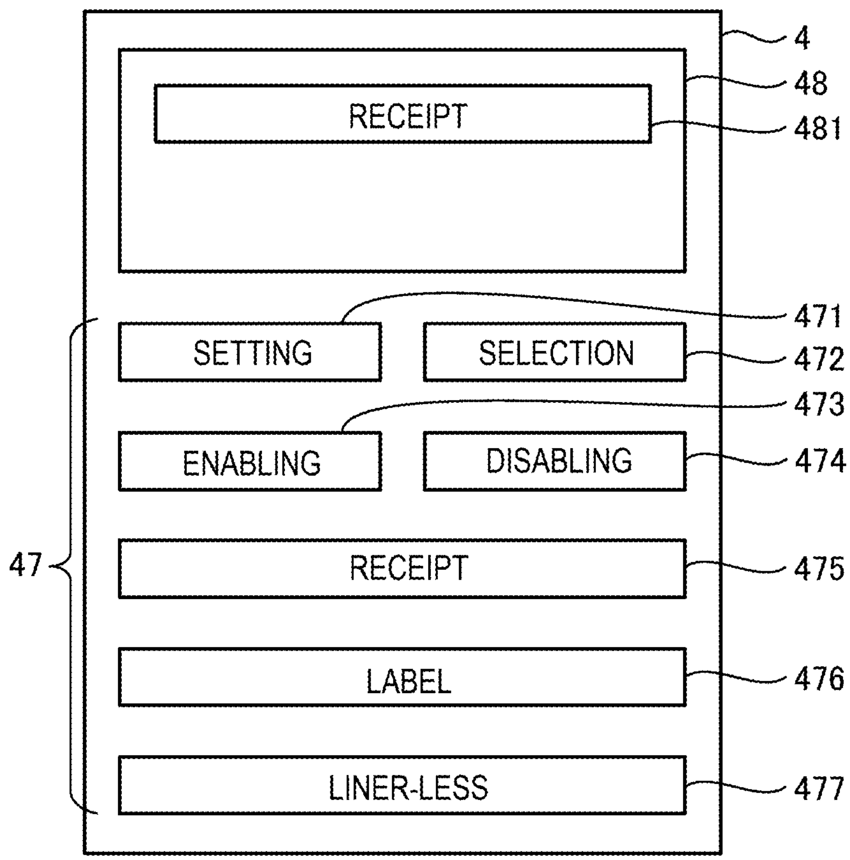
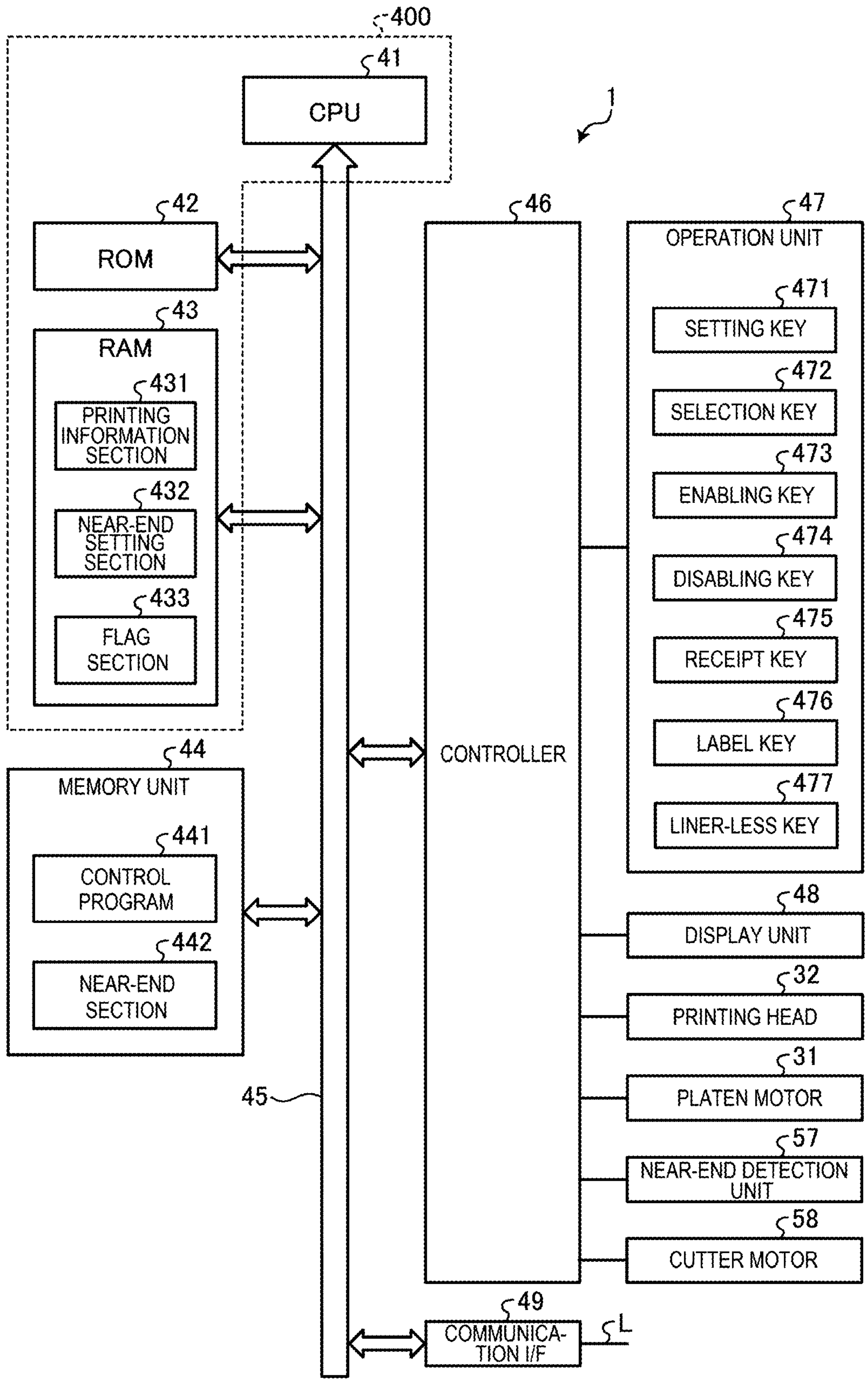

4
48
RECEIPT
481
471
SETTING
SELECTION
472
473
ENABLING
DISABLING
474
47
RECEIPT
475
LABEL
476
LINER-LESS
477

FIG. 5



400
41
CPU
1
42
ROM
43
RAM
431
PRINTING INFORMATION SECTION
432
NEAR-END SETTING SECTION
433
FLAG SECTION
44
MEMORY UNIT
441
CONTROL PROGRAM
442
NEAR-END SECTION
45
46
CONTROLLER
47
OPERATION UNIT
471
SETTING KEY
472
SELECTION KEY
473
ENABLING KEY
474
DISABLING KEY
475
RECEIPT KEY
476
LABEL KEY
477
LINER-LESS KEY
48
DISPLAY UNIT
32
PRINTING HEAD
31
PLATEN MOTOR
57
NEAR-END DETECTION UNIT
58
CUTTER MOTOR
49
COMMUNICA-TION I/F
L

FIG. 6

442

4421 4422

PAPER TYPE SECTION	NEAR-END SETTING SECTION
RECEIPT ROLL PAPER (PAa)	ENABLED
LABEL ROLL PAPER (PBa)	DISABLED
LINER-LESS ROLL PAPER (PCa)	DISABLED

FIG. 7

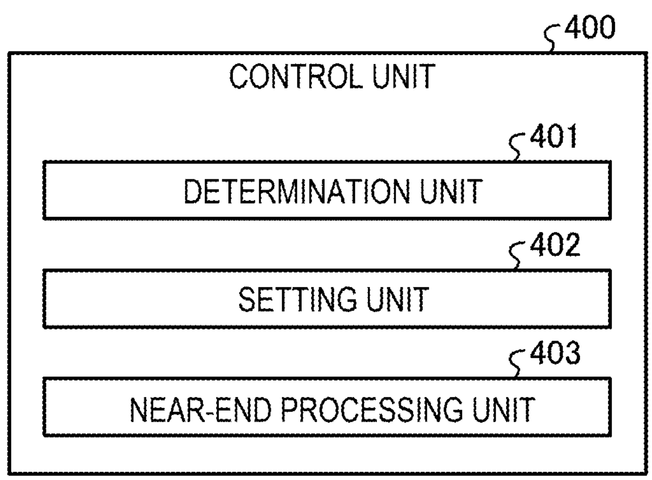


FIG. 8

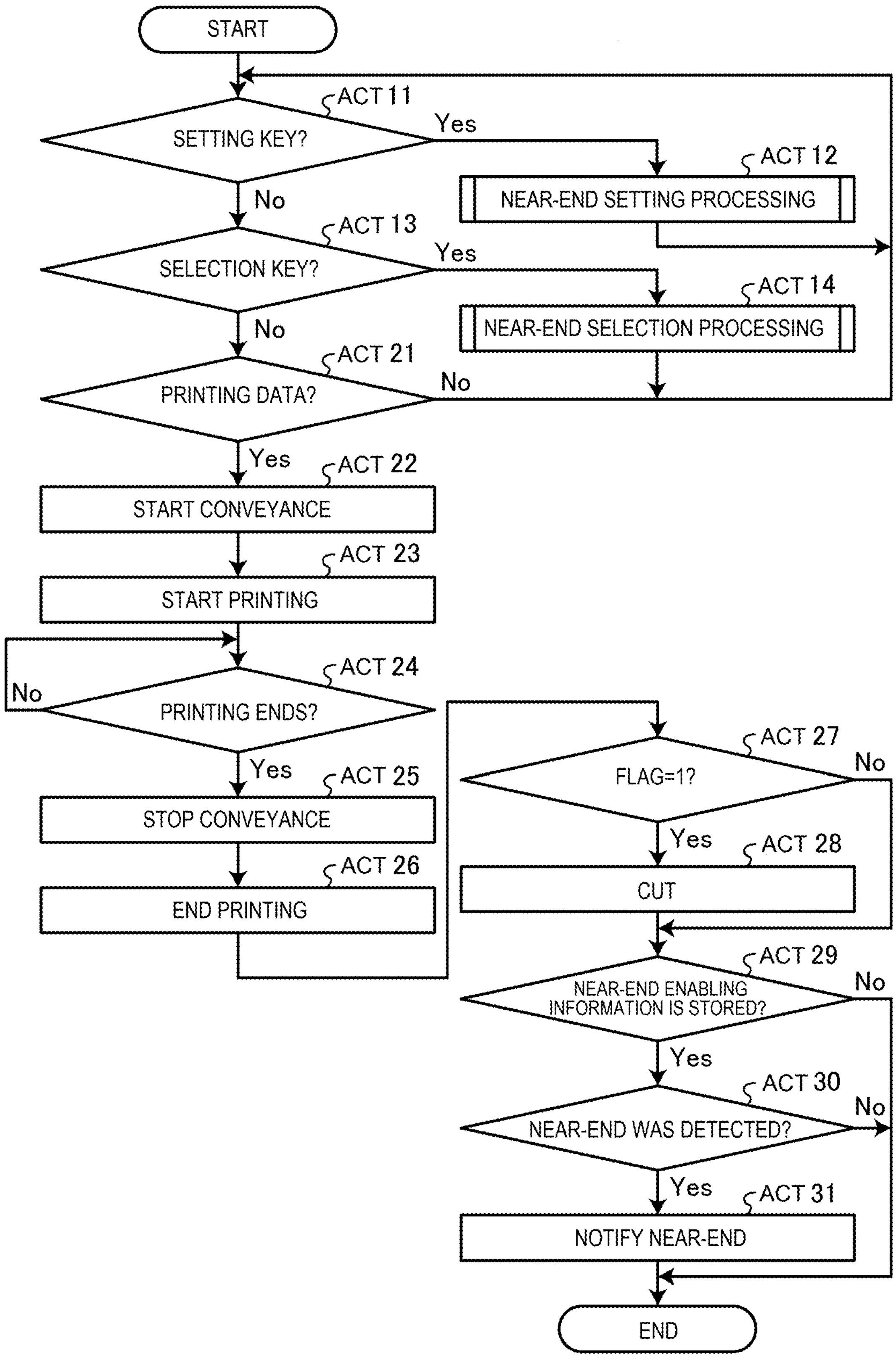

START
ACT 11
SETTING KEY?
Yes
No
ACT 12
NEAR-END SETTING PROCESSING
ACT 13
SELECTION KEY?
Yes
No
ACT 14
NEAR-END SELECTION PROCESSING
ACT 21
PRINTING DATA?
No
Yes
ACT 22
START CONVEYANCE
ACT 23
START PRINTING
ACT 24
PRINTING ENDS?
No
Yes
ACT 25
STOP CONVEYANCE
ACT 26
END PRINTING
ACT 27
FLAG=1?
No
Yes
ACT 28
CUT
ACT 29
NEAR-END ENABLING INFORMATION IS STORED?
No
Yes
ACT 30
NEAR-END WAS DETECTED?
No
Yes
ACT 31
NOTIFY NEAR-END
END

FIG. 9

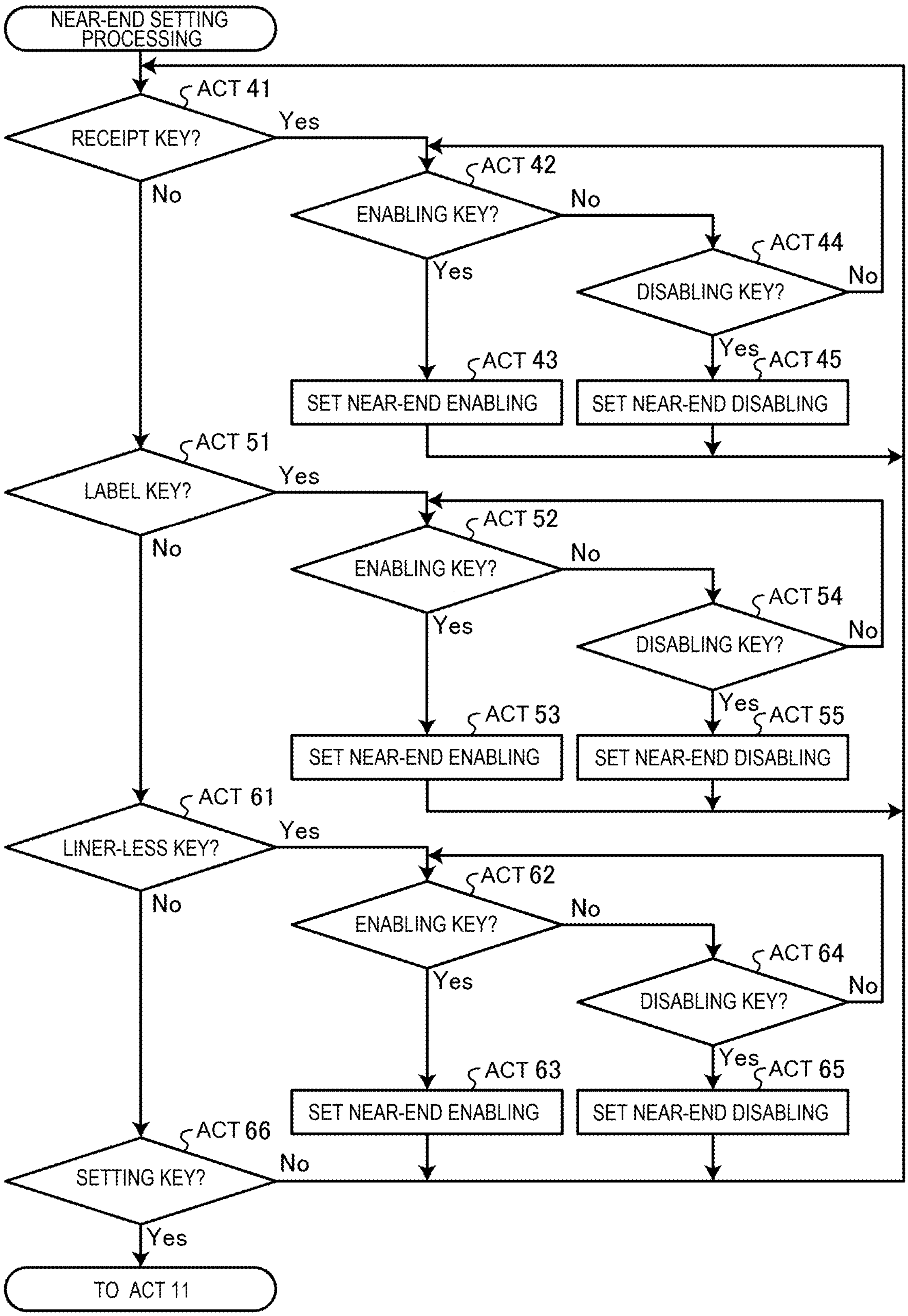
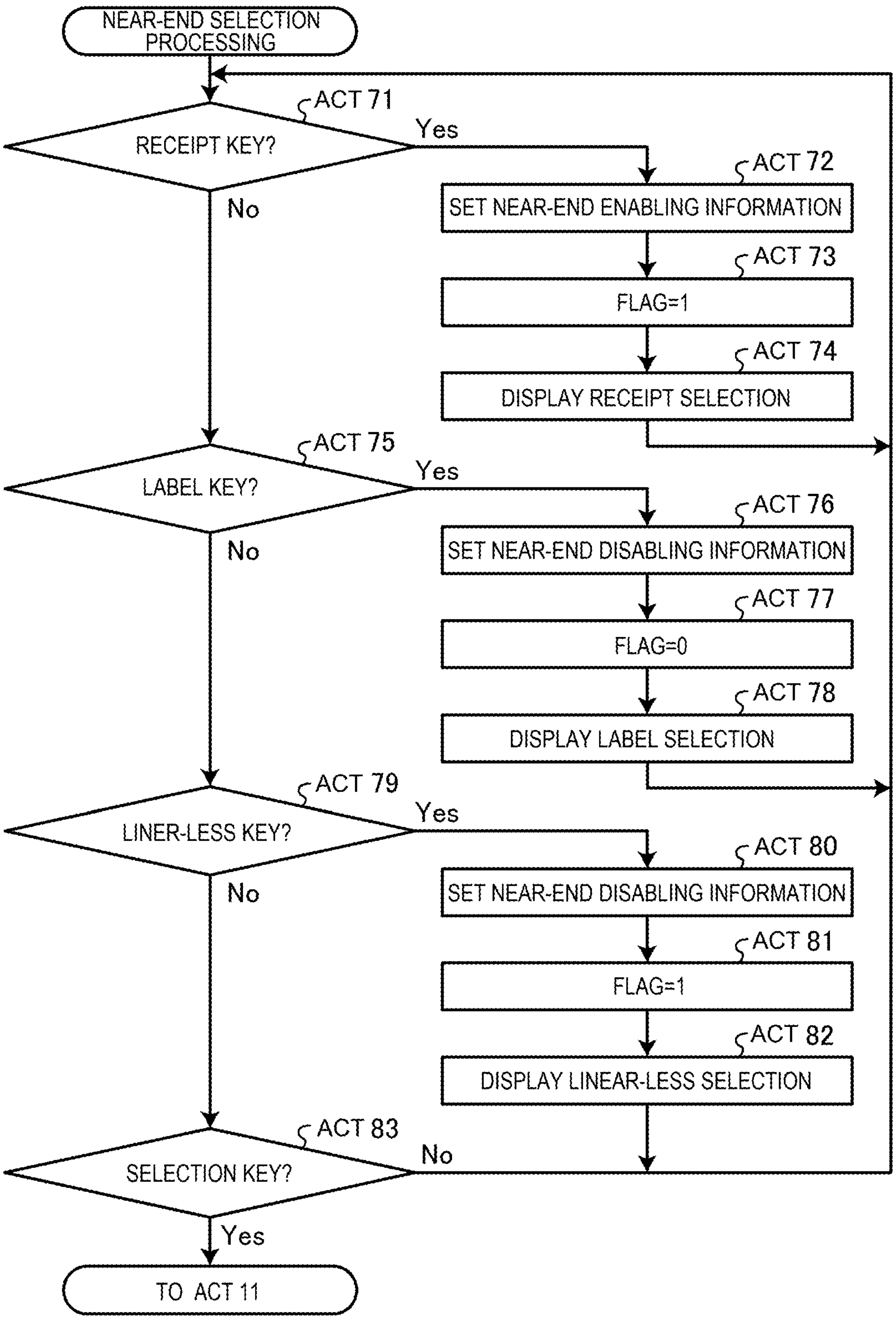

NEAR-END SETTING PROCESSING
ACT 41
RECEIPT KEY?
Yes
No
ACT 42
ENABLING KEY?
No
Yes
ACT 44
DISABLING KEY?
No
Yes
ACT 43
SET NEAR-END ENABLING
ACT 45
SET NEAR-END DISABLING
ACT 51
LABEL KEY?
Yes
No
ACT 52
ENABLING KEY?
No
Yes
ACT 54
DISABLING KEY?
No
Yes
ACT 53
SET NEAR-END ENABLING
ACT 55
SET NEAR-END DISABLING
ACT 61
LINER-LESS KEY?
Yes
No
ACT 62
ENABLING KEY?
No
Yes
ACT 64
DISABLING KEY?
No
Yes
ACT 63
SET NEAR-END ENABLING
ACT 65
SET NEAR-END DISABLING
ACT 66
SETTING KEY?
No
Yes
TO ACT 11

FIG. 10



NEAR-END SELECTION PROCESSING
ACT 71
RECEIPT KEY?
Yes
No
ACT 72
SET NEAR-END ENABLING INFORMATION
ACT 73
FLAG=1
ACT 74
DISPLAY RECEIPT SELECTION
ACT 75
LABEL KEY?
Yes
No
ACT 76
SET NEAR-END DISABLING INFORMATION
ACT 77
FLAG=0
ACT 78
DISPLAY LABEL SELECTION
ACT 79
LINER-LESS KEY?
Yes
No
ACT 80
SET NEAR-END DISABLING INFORMATION
ACT 81
FLAG=1
ACT 82
DISPLAY LINEAR-LESS SELECTION
ACT 83
SELECTION KEY?
No
Yes
TO ACT 11

FIG. 11

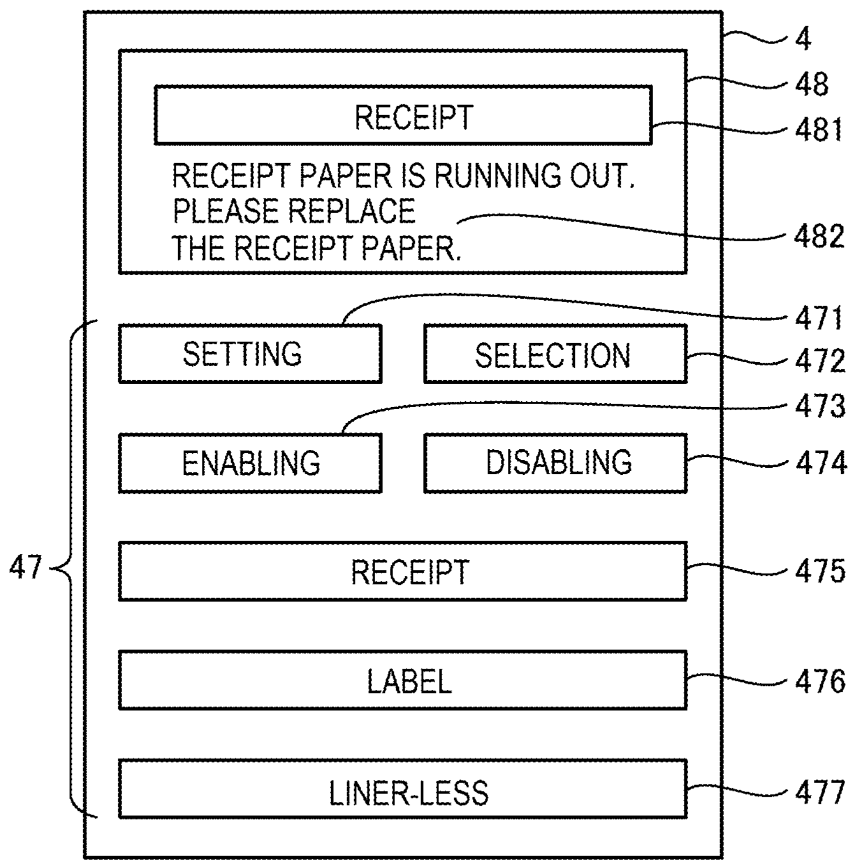
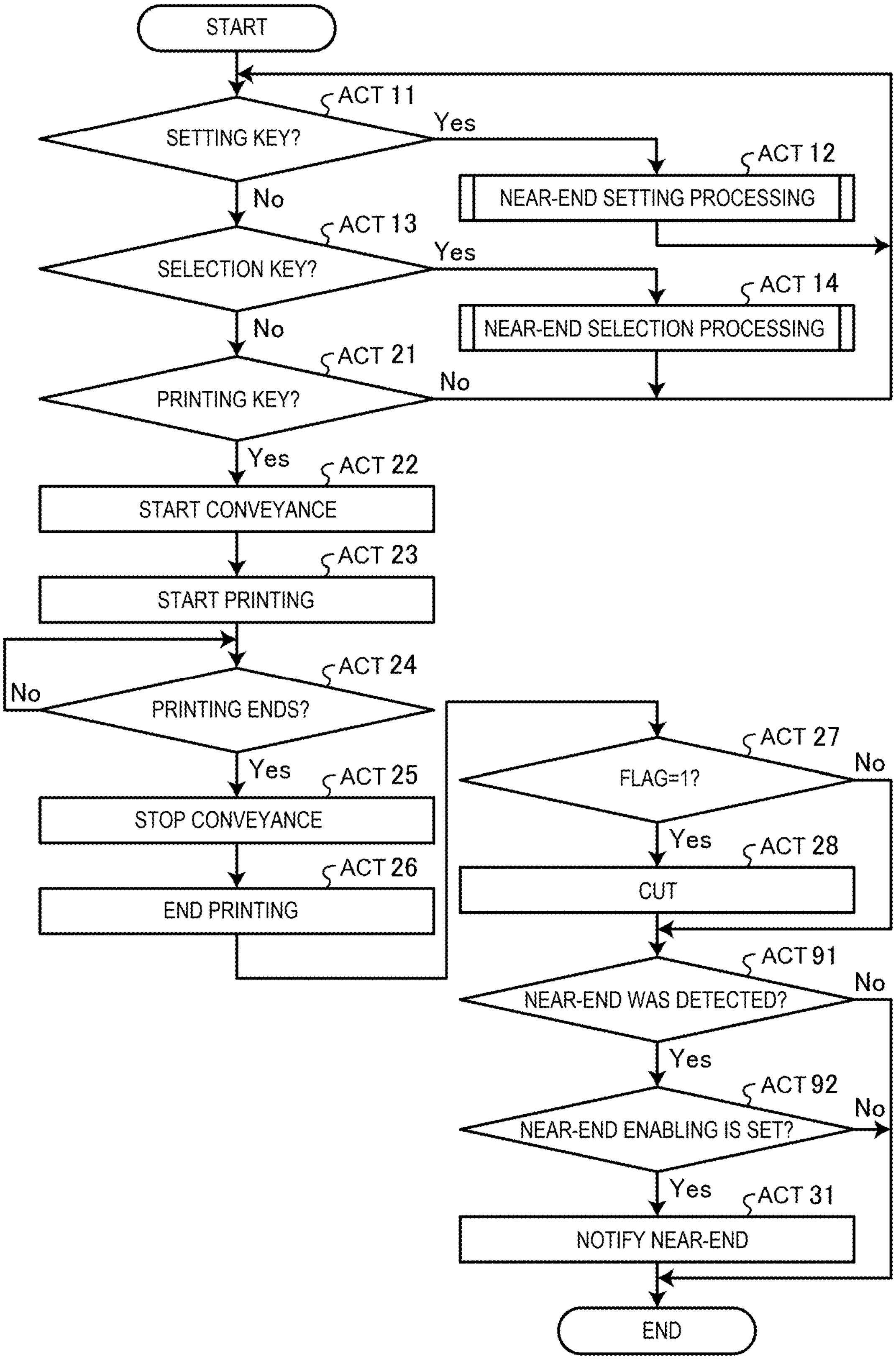

4
48
RECEIPT
481
RECEIPT PAPER IS RUNNING OUT.
PLEASE REPLACE
THE RECEIPT PAPER.
482
471
SETTING
SELECTION
472
473
ENABLING
DISABLING
474
47
RECEIPT
475
LABEL
476
LINER-LESS
477

FIG. 12



START
ACT 11
SETTING KEY?
Yes
No
ACT 12
NEAR-END SETTING PROCESSING
ACT 13
SELECTION KEY?
Yes
No
ACT 14
NEAR-END SELECTION PROCESSING
ACT 21
PRINTING KEY?
No
Yes
ACT 22
START CONVEYANCE
ACT 23
START PRINTING
ACT 24
PRINTING ENDS?
No
Yes
ACT 25
STOP CONVEYANCE
ACT 26
END PRINTING
ACT 27
FLAG=1?
No
Yes
ACT 28
CUT
ACT 91
NEAR-END WAS DETECTED?
No
Yes
ACT 92
NEAR-END ENABLING IS SET?
No
Yes
ACT 31
NOTIFY NEAR-END
END

PRINTER AND METHOD

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-030898, filed on Mar. 1, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to a printer and a method.

BACKGROUND

[0003] For example, there are a receipt printer that prints and dispenses a receipt and a label printer that prints and dispenses a label, a liner-less label, and the like. The receipt printer is installed, together with a point of sale (POS) terminal or the like, in a settlement area for settling a commodity to be sold in a store such as a supermarket, a volume retail store, or a convenience store and dispenses a receipt on which commodity information and settlement information of a commodity subjected to settlement processing by the POS terminal (a commodity purchased by a customer) are printed. The label printer is installed mainly in a back office of a store and prints information concerning a commodity (a commodity name, a commodity price, a best-before date, and the like) on label paper or liner-less label paper and dispenses a label. The dispensed label is stuck to the commodity.

[0004] Incidentally, in recent years, there is a printer functioning as both of a receipt printer and a label printer. That is, if receipt paper (roll paper) is set in the printer, the printer functions as the receipt printer that dispenses a receipt and, if a label or a liner-less label (both of which are roll paper) is set in the printer, the printer functions as the label printer.

[0005] If the printer is used as the receipt printer, operation for setting use of the receipt paper and setting a function of notifying near-end to be enabled is performed. If the printer is used as the label printer, operation for setting use of the label paper or the liner-less paper label and setting the function of notifying the near-end to be disabled is performed. In this way, paper used in the printer and enabling and disabling of near-end are separately set.

DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is an exterior perspective view of a printer in an embodiment;

[0007] FIG. **2** is a perspective view illustrating a configuration of an inside of the opened printer;

[0008] FIG. **3** is an explanatory diagram illustrating internal structure of the printer;

[0009] FIG. **4** is an explanatory diagram illustrating a configuration of a display window;

[0010] FIG. **5** is a block diagram illustrating hardware components of the printer;

[0011] FIG. **6** is a memory map illustrating a memory configuration of a near-end section;

[0012] FIG. **7** is a functional block diagram illustrating functional components of the printer;

[0013] FIG. **8** is a flowchart illustrating a flow of control processing for the printer;

[0014] FIG. **9** is a flowchart illustrating a flow of near-end setting processing;

[0015] FIG. **10** is a flowchart illustrating a flow of near-end selection processing;

[0016] FIG. **11** is an explanatory diagram illustrating a display example of a display unit; and

[0017] FIG. **12** is a flowchart illustrating a flow of control processing in a modification.

DETAILED DESCRIPTION

[0018] An aspect of embodiments is to provide a printer and a method capable of omitting setting of near-end.

[0019] A printer in an embodiment includes: a storage unit configured to store, for each of a plurality of types of roll paper, near-end information indicating whether to notify near-end of the roll paper; a paper set unit in which the roll paper can be set; a determination unit configured to determine a type of the roll paper to be used; a setting unit configured to read near-end information relating to the roll paper, the type of which was determined, from the storage unit and set the near-end information; a near-end detection unit configured to detect the near-end of the roll paper set in the paper set unit; and a near-end processing unit configured to, only if the near-end information indicating that the near-end is notified was set by the setting unit, according to detection of near-end of the roll paper by the near-end detection unit, notify the near-end.

[0020] An embodiment is explained below. FIG. **1** is an exterior perspective view of a printer **1** in the embodiment. The printer **1** in the embodiment is capable of using receipt paper PA, label paper PB, and liner-less paper PC. The printer **1** according to the embodiment is used as, for example, a receipt printer that dispenses a receipt. If the printer **1** is used as the receipt printer, receipt roll paper PAa obtained by winding long paper is set in the printer **1**. The surface of the receipt paper PA is formed to be printable. The printer **1** functioning as the receipt printer is installed adjacent to a POS (Point of Sales) terminal (not illustrated) functioning as an information processing device, for example, in a settlement area of a store. The printer **1** receives commodity information (information such as a commodity code, a commodity a commodity price) of a commodity subjected to settlement processing by the POS terminal and settlement information (information such as a total amount, deposit money, and change, if any, of the commodity subjected to the settlement processing) and prints the commodity information and the settlement information relating to the transaction on receipt draw-out paper PAb drawn out from the receipt roll paper PAa set on the inside of the printer **1**. The printer **1** cuts the printed paper and dispense the cut paper as a receipt PAc.

[0021] The printer **1** is used as, for example, a label printer that dispenses a label. If the printer **1** is used as the label printer, label roll paper PBa obtained by winding a plurality of labels on a long liner in a long direction is set in the printer **1**. The front surface of the label paper PB can be printed. The rear surface of the label paper PB has a glue surface and is peelably bonded to the liner. The printer **1** functioning as the label printer is installed, for example, in a back office of a store. The printer **1** receives printing information from an information processing device such as a PC (Personal Computer) connected to the printer **1** and prints commodity information (information such as a commodity name and a price) relating to the commodity on label draw-out paper PBb drawn out from the label roll paper PBa set on the inside of the printer **1**. The printer **1** dispenses a printed label PBc from a dispensing port.

[0022] The printer **1** is used as, for example, a liner-less printer that dispenses a liner-less label. If the printer **1** is used as the liner-less printer, liner-less roll paper PCa obtained by winding a long liner-less label is set in the printer **1**. The liner-less paper PC is paper used as a long seamless label. The front surface of the liner-less paper PC can be printed. A glue surface is formed on the rear surface of the liner-less paper PC. The printer **1** functioning as the liner-less printer is installed, for example, in a back office of a store. The printer **1** receives printing information from, for example, an information processing device connected to the printer **1** and prints commodity information (information such as a commodity name and a price) concerning the commodity on liner-less draw-out paper PCb drawn out from the liner-less roll paper PCa set on the inside of the printer **1**. The printer **1** dispenses a printed liner-less label PCc from a dispensing port.

[0023] Note that, in the following explanation, the receipt paper PA, the label paper PB, and the liner-less paper PC are sometimes collectively referred to as paper P (for example, if not indicating specific paper). The receipt roll paper PAa, the label roll paper PBa, and the liner-less roll paper PCa are sometimes collectively referred to as roll paper Pa (for example, if not indicating specific roll paper). The receipt draw-out paper PAb, the label draw-out paper PBb, and the liner-less draw-out paper PCb are sometimes collectively referred to as draw-out paper Pb (for example, if not indicating specific paper). The receipt PAc, the label PBc, and the liner-less label PCc are sometimes collectively referred to as paper Pc (for example, if not indicating specific paper).

[0024] As illustrated in FIG. **1**, a housing of the printer **1** is formed by a main body **2** and a lid body **3**. Inside the main body **2**, a power supply unit and a circuit board (both of which are not illustrated), a paper set unit **9** (see FIG. **2**) for setting paper, a platen **33** (see FIG. **2**) serving as a conveying unit that conveys paper, and the like are provided. The lid body **3** is connected to the main body **2** via a hinge **5**. The lid body **3** is attached to be capable of turning in the up-down direction (an arrow Ya direction in FIG. **2**) with respect to the main body **2** with the hinge **5** as a turning fulcrum. The lid body **3** turns upward with respect to the main body **2** to open the inside and turns downward to close the inside of the printer **1**.

[0025] In the lid body **3**, a printing head **32** (see FIG. **2**) and a display window **4** are formed. In the main body **2** or the lid body **3**, a power switch **6** for turning on and off the printer **1** is provided. A paper discharge port **8** is formed in a state in which the lid body **3** is closed on the main body **2**. The paper discharge port **8** discharges (dispenses) printed paper Pc to the outside.

[0026] FIG. **2** is a perspective view illustrating a configuration of the inside of the printer **1** in the state in which the lid body **3** is opened from the main body **2**. As illustrated in FIG. **2**, the lid body **3** is capable of turning in a releasing direction and a closing direction indicated by an arrow Ya with respect to the main body **2**.

[0027] The inside of the main body **2** includes the paper set unit **9** in which the roll paper Pa is set. A method of setting (attaching) the roll paper Pa in the paper set unit **9** may be a dropping method of dropping the roll paper Pa in the paper set unit **9** or may be an axially supporting method of inserting the axis of the roll paper Pa into a shaft (not illustrated) provided in the main body **2** and axially supporting the axis. The roll paper Pa is set in the paper set unit **9** in a state in which the lid body **3** is opened from the main body **2**. In FIG. **2**, a cutter **34** is provided in the main body **2**. The printed paper Pc is cut by moving the cutter **34** upward.

[0028] FIG. **3** is an explanatory diagram illustrating components and positional relations of the units in the state in which the lid body **3** is closed on the main body **2**. As illustrated in FIG. **3**, the roll paper Pa is set in the paper set unit **9**. The printing head **32** is disposed in a position facing the platen **33**. The printing head **32** is in a state of being in press contact with the platen **33**. If the lid body **3** is closed on the main body **2** in a state in which the draw-out paper Pb is drawn out from the roll paper Pa set in the paper set unit **9**, as illustrated in FIG. **3**, the paper P is held by the printing head **32** and the platen **33** are in press contact with each other. If a platen motor **31** is driven in this state, the platen **33** rotates in a direction of an arrow Yc. Then, tension occurs in the paper P held by the printing head **32** and the platen **33**, the roll paper Pa set in the paper set unit **9** rotates in an arrow Yd direction, and the draw-out paper Pb drawn out from the roll paper Pa moves in an arrow Ye direction. The paper P is conveyed in this way.

[0029] The printing head **32** is, for example, a thermal head and applies heat to the conveyed paper P and performs printing. The paper P is, for example, heat sensitive paper. By applying heat to the paper P, characters and figures are printed on the surface (the surface in contact with the printing head **32**) of the paper P. The paper Pc for which the printing is finished is discharged from the paper discharge port **8**. At that time, the receipt paper PA and the liner-less paper PC are cut by the cutter **34** and dispensed from the paper discharge port **8** as the receipt PAc and the liner-les label PCc. The label paper PB is dispensed from the paper discharge port **8** without being cut by the cutter **34**. The receipt paper PA and the liner-less paper PC are cut if a cutter motor **58** is driven to move the cutter **34** in the up-down direction.

[0030] As illustrated in FIG. **3**, a near-end detection unit **57** is provided in the printer **1**. The near-end detection unit **57** is attached to the lid body **3**. If the lid body **3** is closed on the main body **2**, a finger touch section **571** comes into contact with the outer circumferential potion of the roll paper Pa. If the draw-out paper Pb is drawn out from the roll paper Pa set in the paper set unit **9** and the diameter of the roll paper Pa decreases, the finger touch section **571** moves in a direction of an arrow Yb while keeping a state of contact with the outer circumferential portion. If the finger touch section **571** moves to a predetermined position (that is, the diameter of the roll paper Pa decreases to a predetermined length), the near-end detection unit **57** is adjusted to turn on a mechanical switch. The predetermined position is a position where the paper P is in a near-end state. The near-end refers to a state in which the length to an end (a terminal end on the innermost side) of the paper P is the predetermined length and refers to a state in which the paper P is replaced. If the mechanical switch is turned on, the near-end detection unit **57** detects the near-end of the roll paper Pa and outputs a detection signal indicating that the near-end was detected. If the lid body **3** is turned in the releasing direction from the main body **2**, the near-end detection unit **57** separates from the paper P together with the finger touch section **571** according to the turning of the lid body **3**. For that reason, the near-end detection unit **57** is not in the way if the roll paper Pa is set in the paper set unit **9**.

[0031] In the embodiment explained above, if the printer **1** is used as the receipt printer, the near-end of the receipt paper PA (specifically, the receipt roll paper PAa) is notified and, if the printer **1** is used as the label printer, the near-end of the label paper PB (specifically, the label roll paper PBa) is not notified. If the printer **1** is used as the liner-less printer, the near-end of the liner-less paper PC (specifically, the liner-less roll paper PCa) is not notified. For that reason, the printer **1** presets and stores, according to a use form of the printer **1**, near-end enabling information (near-end information) indicating that the near-end is notified and near-end disabling information indicating that the near-end is not notified. That is, in the embodiment, if the printer **1** is used as the receipt printer, the near-end enabling information indicating that the near-end is notified is set in the printer **1**. Specifically, in the embodiment, if the receipt paper PA is set in the printer **1**, the near-end enabling information is set in the printer **1**. In the embodiment, if the printer **1** is used as the label printer or the liner-less printer, the near-end disabling information is set in the printer **1**. Specifically, in the embodiment, if the label paper PB or the liner-less paper PC is set in the printer **1**, the near-end disabling information is set in the printer **1**.

[0032] The printer **1** in which the near-end enabling information is set notifies the near-end. The printer **1** in which the near-end disabling information is set does not notify the near-end.

[0033] The display window **4** is explained below. The printer **1** includes the display window **4** on the upper surface of the lid body **3**. FIG. **4** is an explanatory diagram illustrating a configuration of the display window **4**. As illustrated in FIG. **4**, the display window **4** includes a display unit **48** and an operation unit **47**. The display unit **48** displays information to an operator who operates the printer **1**. The display unit **48** displays a type of the paper P to be used (that is, the roll paper Pa set in the paper set unit **9**). If it is determined that the paper P in use is the receipt paper PA (that is, the roll paper Pa set in the paper set unit **9** is the receipt roll paper PAa), if the near-end detection unit **57** detects near-end, the display unit **48** displays (notifies) the near-end.

[0034] The operation unit **47** includes various keys. The operation unit **47** includes a setting key **471**, a selection key **472**, an enabling key **473**, a disabling key **474**, a receipt key **475**, a label key **476**, and a liner-less key **477**. The receipt key **475**, the label key **476**, and the liner-less key **477** correspond to a designation key for designating a type of the roll paper Pa.

[0035] The setting key **471** is used if near-end information (near-end enabling information or near-end disabling information) is set (stored) in a near-end section **442** (see FIG. **5**) in association with each of types of usable paper P (the receipt paper PA, the label paper PB, and the liner-less paper PC). The enabling key **473** is operated if the near-end enabling information is set. The disabling key **474** is operated if the near-end disabling information is set. In the embodiment, the near-end enabling information is set in association with the receipt paper PA. The near-end disabling information is set in association with the label paper PB and the liner-less paper PC. The selection key **472** is used if the near-end information stored (set) in association with the paper P is set in the printer **1**. Specifically, if the receipt key **475** is operated after the selection key **472** is operated, the near-end information (in the embodiment, the near-end enabling information) stored in association with the receipt paper PA is invoked and set in the printer **1**. If the label key **476** is operated after the selection key **472** is operated, the near-end information (in the embodiment, the near-end disabling information) stored in association with the label paper PB is invoked and set in the printer **1**. If the liner-less key **477** is operated after the selection key **472** is operated, the near-end information (in the embodiment, the near-end disabling information) stored in association with the liner-less paper PC is invoked and set in the printer **1**.

[0036] After operating the selection key **472**, the operator of the printer **1** operates a key (any one of the receipt key **475**, the label key **476**, and the liner-less key **477**) corresponding to a type of the roll paper Pa set in the paper set unit **9** to designate a type of the paper P. The display unit **48** displays characters **481** indicating the designated paper P according to a type of the key operated after the selection key **472** is operated. Specifically, if the receipt key **475** is operated after the selection key **472** is operated, the characters **481** of "receipt" are displayed assuming that the receipt paper PA was designated. If the label key **476** is operated after the selection key **472** is operated, the characters **481** of "label" are displayed assuming that the label paper PB was designated. If the liner-less key **477** is operated after the selection key **472** is operated, the characters **481** of "liner-less" is displayed assuming that the liner-less paper PC was designated. In the example illustrated in FIG. **4**, "receipt" is displayed.

[0037] Hardware of the printer **1** is explained below. FIG. **5** is a block diagram illustrating hardware components of the printer **1**. As illustrated in FIG. **5**, the printer **1** includes a CPU (Central Processing Unit) **41** that is an example of a processor, a ROM (Read Only Memory) **42**, a RAM (Random Access Memory) **43**, and a memory unit **44**. The CPU **41** is a control entity. The ROM **42** stores various programs. Programs and various data are loaded in the RAM **43**. The memory unit **44** stores various programs. The CPU **41**, the ROM **42**, the RAM **43**, and the memory unit **44** are connected to one another via a bus **45**. The CPU **41**, the ROM **42**, and the RAM **43** configure a control unit **400**. That is, the CPU **41** operates according to a control program stored in the ROM **42** or the memory unit **44** and loaded in the RAM **43**, whereby the control unit **400** executes control processing for the printer **1** explained below.

[0038] The RAM **43** includes a printing information section **431**, a near-end setting section **432**, and a flag section **433**. The printing information section **431** stores printing information received from an information processing device. The near-end setting section **432** stores near-end information read from a near-end section **422** explained below. The flag section **433** stores flag information corresponding to a type of selected paper P.

[0039] The memory unit **44** is configured by a nonvolatile memory such as an HDD (Hard Disc Drive) or a flash memory in which stored information is retained even if the power is turned off. The memory unit **44** includes a control program section **441** and a near-end section **422** (a storage unit). The control program section **441** stores a control program for controlling the printer **1**. The near-end section **442** stores near-end information in association with the roll paper Pa to be used (the roll paper Pa set in the paper set unit **9**). The near-end section **442** is explained below with reference to FIG. **6**.

[0040] The control unit **400** is connected to the operation unit **47**, the display unit **48**, the printing head **32**, the platen motor **31**, the near-end detection unit **57**, and the cutter motor **58** via the bus **45** and a controller **46**. The operation unit **47** is a key for operating the printer **1** and includes the setting key **471**, the selection key **472**, the enabling key **473**, the disabling key **474**, the receipt key **475**, the label key **476**, and the liner-less key **477**. The display unit **48** displays information to the operator who operates the printer **1**.

[0041] The printing head **32** applies heat to the surface of the paper P to print characters and figures on the surface of the paper P. The platen motor **31** drives to rotate the platen **33**. The near-end detection unit **57** detects near-end of the roll paper Pa. If receiving a detection signal from the near-end detection unit **57**, the control unit **400** determines that the roll paper Pa set in the paper set unit **9** is near end. The cutter motor **58** drives to move the cutter **34** up and down.

[0042] The controller **46** receives an instruction from the control unit **400** and controls the operation unit **47** and the display unit **48**. In the following explanation, for convenience of explanation, it is assumed that the control unit **400** performs control performed by the controller **46**.

[0043] The control unit **400** is connected to a communication unit **49** via the bus **45**. The communication unit **49** is communicably connected to, via a communication line L such as a LAN (Local Area Network), an information processing device that transmits printing information.

[0044] Subsequently, the near-end section **442** is explained. FIG. **6** is a memory map illustrating a memory configuration of the near-end section **442**. The near-end section **442** stores near-end information set for each of types of the paper P (specifically, the roll paper Pa) in association with a plurality of types of the paper P (specifically, the roll paper Pa) usable in the printer **1**. As illustrated in FIG. **6**, the near-end section **442** includes a paper type section **4421** and a near-end setting section **4422**. The paper type section **4421** stores information concerning the paper P (specifically, the roll paper Pa) that can be used in the printer **1** (can be set in the paper set unit **9**). In the embodiment, as the information concerning the paper P that can be used in the printer **1** (can be set in the paper set unit **9**), the paper type section **4421** stores information for specifying the receipt paper PA (specifically, the receipt roll paper PAa), information for specifying the label paper PB (specifically, the label roll paper PBa), and information for specifying the liner-less paper PC (specifically, the liner-less roll paper PCa). The near-end setting section **4422** stores near-end information in association with the paper P (specifically, the roll paper Pa) stored in the near-end section **442**. In the embodiment, the near-end setting section **4422** stores, in association with the receipt paper PA (specifically, the receipt roll paper PAa), near-end enabling information indicating that near-end is notified. The near-end setting section **4422** stores, in association with the label paper PB (specifically, the label roll paper PBa), near-end disabling information indicating that near-end is not notified. The near-end setting section **4422** stores, in association with the liner-less paper PC (specifically, the liner-less roll paper PCa), near-end disabling information indicating that near-end is not notified.

[0045] Functional components of the printer **1** are explained below. FIG. **7** is a functional block diagram illustrating the functional components of the printer **1**. The control unit **400** of the printer **1** follows a control program stored in the ROM **42** or the control program section **441** of the memory unit **44** to function as a determination unit **401**, a setting unit **402**, and a near-end processing unit **403**.

[0046] The determination unit **401** determines a type of the roll paper Pa to be used. Specifically, the determination unit **401** determines, based on input information indicating a type of the roll paper Pa, a type of the roll paper Pa to be used. In the embodiment, if any one key among the receipt key **475**, the label key **476**, and the liner-less key **477** is operated, since a key signal indicating the operated key is input, the determination unit **401** determines a type of the operated key (that is, a type of the roll paper Pa to be used) based on the input key signal.

[0047] The setting unit **402** reads near-end information relating to the roll paper Pa, the type of which was determined, from the near-end section **442** and sets the near-end information. Specifically, the setting unit **402** reads the near-end information relating to the roll paper Pa, the type of which was determined, from the near-end section **442** and stores the near-end information in the near-end setting section **432**. In the embodiment, the setting unit **402** determines, based on the input key signal, which key among the receipt key **475**, the label key **476**, and the liner-less key **477** was operated, determines the roll paper Pa to be used, reads near-end information corresponding to the determined roll paper Pa, and stores the near-end information in the near-end setting section **432**.

[0048] Only if the near-end enabling information indicating that near-end is notified was set by the setting unit **402**, according to detection of near-end of the roll paper Pa by the near-end detection unit **57**, the near-end processing unit **403** notifies the near-end. In the embodiment, the near-end enabling information is set for the receipt paper PA and the near-end disabling information is set (that is, the near-end enabling information is not set) for the label paper PB and the liner-less paper PC. Therefore, if near-end of the receipt roll paper PAa is detected by the near-end detection unit **57**, the near-end processing unit **403** notifies the near-end. If near-end of the label roll paper PBa is detected by the near-end detection unit **57** and if near-end of the liner-less roll paper PCa is detected by the near-end detection unit **57**, the near-end processing unit **403** does not notify the near-end.

[0049] Control of the printer **1** is explained below. FIG. **8** is a flowchart illustrating a flow of control processing for the printer **1**. As illustrated in FIG. **8**, the control unit **400** of the printer **1** determines whether the setting key **471** was operated (Act **11**). If determining that the setting key **471** was operated (Yes in Act **11**), the control unit **400** executes near-end setting processing (Act **12**).

[0050] FIG. **9** is a flowchart illustrating the near-end setting processing executed in Act **12**. As illustrated in FIG. **9**, the control unit **400** determines whether the receipt key **475** was operated (Act **41**). If determining that the receipt key **475** was operated (Yes in Act **41**), subsequently, the control unit **400** determines whether the enabling key **473** was operated (Act **42**). If determining that the enabling key **473** was operated (Yes in Act **42**), the control unit **400** sets (stores), in the near-end setting section **4422** corresponding to information indicating the receipt roll paper PAa stored in the paper type section **4421**, "near-end enabling information" that is near-end information indicating that, if the near-end detection unit **57** detected near-end, the near-end is notified (Act **43**). Then, the control unit **400** returns to Act **41**.

[0051] If determining that the enabling key **473** was not operated (No in Act **42**), the control unit **400** determines whether the disabling key **474** was operated (Act **44**). If determining that the disabling key **474** was operated (Yes in Act **44**), the control unit **400** sets (stores), in the near-end setting section **4422** corresponding to the information indicating the receipt roll paper PAa stored in the paper type section **4421**, "near-end disabling information" that is near-end information indicating that it is not determined whether near-end was detected by the near-end detection unit **57** (Act **45**). Then, the control unit **400** returns to Act **41**. In the embodiment, since near-end is notified if the receipt roll paper PAa is set, the "near-end enabling information" is set in the near-end setting section **4422** corresponding to the information indicating the receipt roll paper PAa. If determining that the disabling key **474** was not operated (No in Act **44**), the control unit **400** returns to Act **42**.

[0052] If determining that the receipt key **475** was not operated (No in Act **41**), the control unit **400** determines whether the label key **476** was operated (Act **51**). If determining that the label key **476** was operated (Yes in Act **51**), subsequently, the control unit **400** determines whether the enabling key **473** was operated (Act **52**). If determining that the enabling key **473** was operated (Yes in Act **52**), the control unit **400** sets (stores), in the near-end setting section **4422** corresponding to information indicating the label roll paper PBa stored in the paper type section **4421**, "near-end enabling information" that is near-end information indicating that, if the near-end detection unit **57** detected near-end, the near-end is notified (Act **53**). Then, the control unit **400** returns to Act **41**.

[0053] If determining that the enabling key **473** was not operated (No in Act **52**), the control unit **400** determines whether the disabling key **474** was operated (Act **54**). If determining that the disabling key **474** was operated (Yes in Act **54**), the control unit **400** sets (stores), in the near-end setting section **4422** corresponding to information indicating the label roll paper PBa stored in the paper type section **4421**, "near-end disabling information" that is near-end information indicating that it is not determined whether near-end was detected by the near-end detection unit **57** (Act **55**). Then, the control unit **400** returns to Act **41**. In the embodiment, since near-end is not notified if the label roll paper PBa is set, the "near-end disabling information" is set in the near-end setting section **4422** corresponding to the information indicating the label roll paper PBa. If determining that the disabling key **474** was not operated (No in Act **54**), the control unit **400** returns to Act **52**.

[0054] If determining that the label key **476** was not operated (No in Act **51**), the control unit **400** determines whether the liner-less key **477** was operated (Act **61**). If determining that the liner-less key **477** was operated (Yes in Act **61**), subsequently, the control unit **400** determines whether the enabling key **473** was operated (Act **62**). If determining that the enabling key **473** was operated (Yes in Act **62**), the control unit **400** sets (stores), in the near-end setting section **4422** corresponding to information indicating the liner-less roll paper PCa stored in the paper type section **4421**, "near-end enabling information" that is near-end information indicating that, if the near-end detection unit **57** detected near-end, the near-end is notified (Act **63**). Then, the control unit **400** returns to Act **41**.

[0055] If determining that the enabling key **473** was not operated (No in Act **62**), the control unit **400** determines whether the disabling key **474** was operated (Act **64**). If determining that the disabling key **474** was operated (Yes in Act **64**), the control unit **400** sets (stores), in the near-end setting section **4422** corresponding to the information indicating the liner-less roll paper PCa stored in the paper type section **4421**, "near-end disabling information" that is near-end information indicating that it is not determined whether near-end was detected by the near-end detection unit **57** (Act **65**). Then, the control unit **400** returns to Act **41**. In the embodiment, since near-end is not notified if the liner-less roll paper PCa is set, the "near-end disabling information" is set in the near-end setting section **4422** corresponding to the information indicating the liner-less roll paper PCa. If determining that the disabling key **474** was not operated (No in Act **64**), the control unit **400** returns to Act **62**.

[0056] If determining that the liner-less key **477** was not operated (No in Act **61**), the control unit **400** determines whether the setting key **471** was operated again (Act **66**). If determining that the setting key **471** was operated (Yes in Act **66**), the control unit **400** returns to Act **11**. If determining that the setting key **471** was not operated (No in Act **66**), the control unit **400** returns to Act **41**.

[0057] Referring back to FIG. **8**, if determining that the setting key **471** was not operated (No in Act **11**), the control unit **400** determines whether the selection key **472** was operated (Act **13**). If determining that the selection key **472** was operated (Yes in Act **13**), the control unit **400** executes near-end selection processing (Act **14**).

[0058] FIG. **10** is a flowchart illustrating the near-end selection processing executed in Act **14**. If the operator of the printer **1** set the receipt roll paper PAa in the paper set unit **9**, the operator of the printer **1** operates the receipt key **475** subsequently to the operation of the selection key **472**. If the operator set the label roll paper PBa in the paper set unit **9**, the operator operates the label key **476** subsequently to the operation of the selection key **472**. If the operator set the liner-less roll paper PCa in the paper set unit **9**, the operator operates the liner-less key **477** subsequently to the operation of the selection key **472**.

[0059] As illustrated in FIG. **10**, the determination unit **401** determines whether the receipt key **475** was operated (Act **71**). If it is determined that the receipt key **475** was operated (Yes in Act **71**), the setting unit **402** reads near-end information (in the embodiment, "near-end enabling information") stored in the near-end setting section **4422** corresponding to the receipt roll paper PAa stored in the paper type section **4421** and stores the near-end information in the near-end setting section **432** (Act **72**). The control unit **400** stores, in the flag section **433**, flag information "1" indicating that the control unit **400** causes the cutter **34** to operate (Act **73**). The control unit **400** displays, on the display unit **48**, the characters **481** (in the embodiment, "receipt") indicating that the receipt roll paper PAa was set in the paper set unit **9** (Act **74**). Then, the control unit **400** returns to Act **71**.

[0060] If determining that the receipt key **475** was not operated (No in Act **71**), the determination unit **401** determines whether the label key **476** was operated (Act **75**). If it is determined that the label key **476** was operated (Yes in Act **75**), the setting unit **402** reads near-end information (in the embodiment, "near-end disabling information") stored in the near-end setting section **4422** corresponding to the label roll paper PBa stored in the paper type section **4421** and stores the near-end information in the near-end setting section **432** (Act **76**). The control unit **400** stores, in the flag section **433**, flag information "0" indicating that the control unit **400** does not cause the cutter **34** to operate (Act **77**). The control unit **400** displays, on the display unit **48**, the characters **481** (in the embodiment, "label") indicating that the label roll paper PBa was set in the paper set unit **9** (Act **78**). Then, the control unit **400** returns to Act **71**.

[0061] If determining that the label key **476** was not operated (No in Act **75**), the determination unit **401** determines whether the liner-less key **477** was operated (Act **79**). If it is determined that the liner-less key **477** was operated (Yes in Act **79**), the setting unit **402** reads near-end information (in the embodiment, "near-end disabling information") stored in the near-ed setting section **4422** corresponding to the liner-less roll paper PCa stored in the paper type section **4421** and stores the near-end information in the near-end setting section **432** (Act **80**). The control unit **400** stores, in the flag section **433**, flag information "1" indicating that the control unit **400** causes the cutter **34** to operate (Act **81**). The control unit **400** displays, on the display unit **48**, the characters **481** (in the embodiment, "liner-less") indicating that the liner-less roll paper PCa was set in the paper set unit **9** (Act **82**). Then, the control unit **400** returns to Act **71**.

[0062] If determining that the liner-less key **477** was not operated (No in Act **79**), the control unit **400** determines whether the selection key **472** was operated again (Act **83**). If determining that the selection key **472** was operated (Yes in Act **83**), the control unit **400** returns to Act **11**. If determining that the selection key **472** was not operated (No in Act **83**), the control unit **400** returns to Act **71**.

[0063] Referring back to FIG. **8**, if determining that the selection key **472** was not operated (No in Act **13**), the control unit **400** determines whether printing data was received from the information processing device (Act **21**). If determining that printing data was not received (No in Act **21**), the control unit **400** returns to Act **11**. If determining that printing data was received (Yes in Act **21**), the control unit **400** drives the platen motor **31** to rotate the platen **33** and starts conveyance of the paper P (Act **22**). Then, the control unit **400** drives the printing head **32** to start printing of information on the surface of the paper P (Act **23**).

[0064] Subsequently, the control unit **400** determines whether the printing of the received printing information ended (all the received printing information was printed) (Act **24**). The control unit **400** continues the conveyance of the paper P and the printing until the printing ends (No in Act **24**). If determining that the printing of the printing information ended (Yes in Act **24**), the control unit **400** stops the rotation of the platen **33** and stops the conveyance of the paper P (Act **25**). Then, the control unit **400** ends the printing by the printing head **32** (Act **26**). The paper Pc printed in this way is discharged from the paper discharge port **8**.

[0065] Subsequently, the control unit **400** checks flag information stored in the flag section **433** and determines whether flag information "1" is stored in the flag section **433** (Act **27**). If determining that the flag information "1" is stored (Yes in Act **27**), the control unit **400** drives the cutter **34** to cut the paper Pc discharged (dispensed) from the paper discharge port **8** (Act **28**). If the flag information "1" is not stored in the flag section **433** (that is, flag information "0" is stored) (No in Act **27**), the control unit **400** does not drive the cutter **34**. That is, the control unit **400** does not execute processing in Act **28**.

[0066] Subsequently, the control unit **400** determines whether near-end enabling information is stored in the near-end setting section **432** (Act **29**). If determining that the near-end enabling information is stored (Yes in Act **29**), subsequently, the control unit **400** determines whether the near-end detection unit **57** detected near-end of the roll paper Pa (Act **30**). If it is determined that near-end was detected (Yes in Act **30**), the near-end processing unit **403** displays (notifies), on the display unit **48**, near-end detection information indicating that the paper P is near end (Act **31**).

[0067] FIG. **11** illustrates the display unit **48** that displays, in Act **31**, near-end detection information **482** indicating that the receipt paper PA (the receipt roll paper PAa) is near end. As illustrated in FIG. **11**, the display unit **48** displays the near-end detection information **482** indicating that the receipt roll paper PAa is near end. In an example illustrated in FIG. **11**, as the near-end detection information **482**, for example, "Receipt paper is running out. Please replace the receipt paper." is displayed. The control unit **400** ends the processing and returns to Act **11**.

[0068] Referring back to FIG. **8**, if determining that the near-end enabling information is not stored in the near-end setting section **432** (that is, near-end disabling information is stored) (No in Act **29**), the control unit **400** ends the processing without determining (the determination in Act **30**) whether the near-end detection unit **57** detected near-end and returns to Act **11**. That is, if determining No in Act **29**, since the control unit **400** does not perform the determination in Act **30**, the control unit **400** does not perform the processing in Act **31** (processing for displaying (notifying) the near-end detection information on the display unit **48**). If determining that near-end was not detected (No in Act **30**), the control unit **400** ends the processing and returns to Act **11**.

[0069] According to the embodiment explained above, if a type of the roll paper Pa was designated by the designation key, near-end information stored in the near-end section **442** and corresponding to the designated roll paper Pa is stored in the near-end setting section **432**, on condition that near-end enabling information indicating that near-end is notified is stored in the near-end setting section **432**, it is determined whether the near-end detection unit **57** detected near-end, and, if it is determined that the near-end detection unit **57** detected near-end, near-end detection information indicating that the roll paper Pa is near end is displayed on the display unit **48**. Conversely, if near-end disabling information indicating that near-end is not notified is stored in the near-end setting section **432**, it is not determined whether the near-end detection unit **57** detected near-end and near-end detection information indicating that the roll paper Pa is near end is not displayed on the display unit **48**.

[0070] A modification of the printer **1** is explained below. In the embodiment, if near-end enabling information indicating that near-end is notified is stored in the near-end setting section **432**, it is determined whether the near-end detection unit **57** detected near-end of the paper P (the determination in Act **30** is performed in the case of Yes in the determination in Act **29**). If near-end disabling information indicating that near-end is not detected is stored in the near-end setting section **432**, it is not determined whether the near-end detection unit **57** detected near-end of the paper P (the determination in Act **30** is not executed in the case of No in the determination in Act **29**).

[0071] In the modification of the embodiment, first, it is determined whether the near-end detection unit **57** detected near-end of the paper P. If it is determined that the near-end detection unit **57** detected near-end, subsequently, it is determined whether near-end enabling information indicating that near-end is notified is stored in the near-end setting section **432**. In the explanation of the modification, the same components as the components in the embodiment are denoted by the same reference numerals and signs and explanation of the components is omitted or simplified.

[0072] FIG. **12** is a flowchart illustrating a flow of control processing in the modification. FIG. **12** is a diagram corresponding to FIG. **8** in the embodiment. As illustrated in FIG. **12**, the same processing as the processing in the embodiment is performed in Act **11** to Act **28** and Act **31**. In the modification, the processing in Act **45**, Act **55**, and Act **65** in FIG. **9** in the near-end setting processing (Act **12**) is performed as explained below.

[0073] In the case of the modification, in the case of Yes in the determination in Act **44**, the control unit **400** sets (stores), in the near-end setting section **4422** corresponding to the information indicating the receipt roll paper PAa stored in the paper type section **4421**, "near-end disabling information" that is near-end information indicating that, even if near-end was detected by the near-end detection unit **57**, the near-end is not notified (Act **45**). In the case of Yes in the determination in Act **54**, the control unit **400** sets (stores), in the near-end setting section **4422** corresponding to the information indicating the label roll paper PBa stored in the paper type section **4421**, "near-end disabling information" that is near-end information indicating that, even if near-end was detected by the near-end detection unit **57**, the near-end is not notified (Act **55**). In the case of Yes in the determination in Act **64**, the control unit **400** sets (stores), in the near-end setting section **4422** corresponding to the information indicating the liner-less roll paper PCa stored in the paper type section **4421**, "near-end disabling information" that is near-end information indicating that, even if near-end was detected by the near-end detection unit **57**, the near-end is not notified (Act **65**).

[0074] After executing the processing in Act **28**, the control unit **400** determines whether the near-end detection unit **57** detected near-end of the roll paper Pa (Act **91**). If determining that the near-end detection unit **57** detected near-end (Yes in Act **91**), subsequently, the control unit **400** determines whether near-end enabling information is stored in the near-end setting section **432** (Act **92**). If it is determined that the near-end enabling information is stored (Yes in Act **92**), the near-end processing unit **403** displays (notifies), on the display unit **48**, near-end detection information indicating that the paper P is near end (Act **31**). If it is determined that near-end enabling information is not stored (near-end disabling information is stored) (No in Act **92**), the near-end processing unit **403** does not display the near-end detection information on the display unit **48**. That is, if it is determined that near-end enabling information is not stored (No in Act **92**), the near-end processing unit **403** does not execute the processing in Act **31**. That is, even if the near-end detection unit **57** detects near-end of the roll paper Pa, if near-end disabling information is stored in the near-end setting section **432**, the near-end processing unit **403** does not display the near-end detection information on the display unit **48**.

[0075] According to the modification explained above, if a type of the roll paper Pa is designated by a designation key, near-end information corresponding to the selected roll paper Pa stored in the near-end section **442** is stored in the near-end setting section **432**, on condition that the near-end detection unit **57** detected near-end, it is determined whether near-end enabling information is stored in the near-end setting section **432**, and, if it is determined that near-end enabling information is stored, near-end detection information indicating that the roll paper Pa is near end is displayed on the display unit **48**. Conversely, if near-end disabling information is stored in the near-end setting section **432**, even if the near-end detection unit **57** detects near-end, the near-end detection information indicating that the roll paper Pa is near end is not displayed on the display unit **48**.

[0076] As explained above, the printer **1** in the embodiment and the modification includes the near-end section **442** that stores, for each of a plurality of types of roll paper Pa, near-end information indicating whether near-end of the roll paper Pa is notified, the paper set unit **9** in which the roll paper Pa can be set, the determination unit **401** that determines a type of the roll paper Pa to be used, the setting unit **402** that reads near-end information relating to the roll paper Pa, the type of which was determined, from the near-end section **442** and sets the near-end information, the near-end detection unit **57** that detects near-end of the roll paper Pa set in the paper set unit **9**, and the near-end processing unit **403** that, only if near-end information indicating that near-end is notified was set by the setting unit **402**, according to detection of near-end of the roll paper Pa by the near-end detection unit **57**, notifies the near-end.

[0077] With the printer **1** in the embodiment and the modification explained above, if a type of the roll paper Pa to be used is determined, near-end information relating to the roll paper Pa is read from the near-end section **442** and stored in the near-end setting section **432**, and, if near-end enabling information is set in the near-end setting section **432**, according to detection of near-end of the roll paper Pa by the near-end detection unit **57**, the near-end is notified. For that reason, it is possible to omit setting of near-end performed every time separately from setting of the roll paper Pa in the related art.

[0078] The embodiment is explained above. However, the embodiment is presented as an example and is not intended to limit the scope of the invention. The new embodiment can be implemented in other various forms. Various omissions, substitutions, changes, and combinations can be made without departing from the gist of the invention. The embodiment and modifications of the embodiment are included in the scope and the gist of the invention and included in the inventions described in claims and the scope of equivalents of the inventions.

[0079] For example, in the embodiment and the modification, the near-end detection unit **57** is explained as the mechanical switch. However, not only this, but, for example, the near-end detection unit **57** may be a transmissive or reflective optical sensor.

[0080] In the embodiment and the modification, the near-end enabling information is stored in association with the receipt paper PA and the near-end disabling information is stored in association with the label paper PB and the liner-less paper PC. However, not only this, but the near-end enabling information only has to be stored in association with the paper P for which near-end needs to be notified.

[0081] In the embodiment and the modification, the receipt paper PA and the liner-less paper PC are cut by the cutter **34**. However, the cut of the paper P is not an essential requirement.

[0082] Programs to be executed in the printer **1** in the embodiment and the modification are provided by being recorded in a computer-readable recording medium such as a CD-ROM, a flexible disk (FD), a CD-R, or a DVD (Digital Versatile Disk) as a file of an installable format or an executable format.

[0083] The programs to be executed in the printer in the embodiment and the modification may be configured to be stored on a computer connected to a network such as the Internet and provided by being downloaded through the network. The programs to be executed in the printer **1** in the embodiment and the modification may be configured to be provided or distributed through the network such as the Internet.

[0084] The programs to be executed in the printer **1** in the embodiment and the modification may be configured to be provided by being incorporated in a ROM or the like in advance.

What is claimed is:

1. A printer, comprising:

a storage component configured to store, for each of a plurality of types of roll paper, near-end information indicating whether to notify near-end of the roll paper;

a paper set component in which the roll paper is set;

a determination component configured to determine a type of the roll paper to be used;

a setting component configured to read near-end information relating to the roll paper, the type of which is determined, from the storage component and set the near-end information;

a near-end detection component configured to detect the near-end of the roll paper set in the paper set component; and

a near-end processing component configured to, only if the near-end information indicating that the near-end is notified is set by the setting component, according to detection of near-end of the roll paper by the near-end detection component, notify the near-end.

2. The printer according to claim **1**, wherein

the plurality of types of roll paper comprise receipt roll paper obtained by winding long receipt paper, label roll paper obtained by aligning and winding a plurality of labels in a long direction on a long liner, and liner-less roll paper obtained by winding long liner-less paper,

the storage component stores, in association with the receipt roll paper, near-end information for notifying near-end and stores, in association with the label roll paper and the liner-less roll paper, near-end information for not notifying near-end, and

the setting component sets near-end information indicating that the near-end is notified if the receipt roll paper is determined by the determination component.

3. The printer according to claim **1**, wherein, if near-end information indicating that the near-end is not notified is set by the setting component, the near-end processing component does not determine whether near-end of the roll paper is detected by the near-end detection component and does not notify the near-end.

4. The printer according to claim **1**, wherein, if near-end information indicating that the near-end is not notified is set by the setting component, even if near-end of the roll paper is detected by the near-end detection component, the near-end processing unit does not notify the near-end.

5. The printer according to claim **1**, further comprising a designation key that is operated in order to designate a type of roll paper, wherein

the determination component determines the roll paper for which the designation key operates.

6. The printer according to claim **1**, wherein the near-end information comprises near-end disabling information and near-end enabling information.

7. The printer according to claim **1**, wherein the determination component determines that when a receipt key is operated, the setting component reads near-end information stored in a near-end setting component corresponding to the roll paper stored in the paper set component and stores the near-end information in the near-end setting component.

8. A method for a printer including a storage component configured to store, for each of a plurality of types of roll paper, near-end information indicating whether to notify near-end of the roll paper, a paper set component in which the roll paper can be set, and a near-end detection unit configured to detect the near-end of the roll paper set in the paper set component,

the method comprising:

determining a type of the roll paper to be used;

reading near-end information relating to the roll paper, the type of which is determined, from the storage component and setting the near-end information; and

only if the near-end information indicating that the near-end is notified is set, according to detection of near-end of the roll paper, notifying the near-end.

9. The method to claim **8**, wherein the plurality of types of roll paper comprise receipt roll paper obtained by winding long receipt paper, label roll paper obtained by aligning and winding a plurality of labels in a long direction on a long liner, and liner-less roll paper obtained by winding long liner-less paper, the method further comprising:

storing, in association with the receipt roll paper, near-end information for notifying near-end and storing, in association with the label roll paper and the liner-less roll paper, near-end information for not notifying near-end, and

setting near-end information indicating that the near-end is notified if the receipt roll paper is determined.

10. The method according to claim **8**, further comprising:

if near-end information indicating that the near-end is not notified is set, not determining whether near-end of the roll paper is detected by the near-end detection component and not notifying the near-end.

11. The method according to claim **8**, further comprising:

if near-end information indicating that the near-end is not notified is set, even if near-end of the roll paper is detected, not notifying the near-end.

12. The method according to claim **8**, further comprising:

operating a designation key in order to designate a type of roll paper; and

determining the roll paper for which the designation key operates.

13. The method according to claim **8**, further comprising:

determining that when a receipt key is operated, the reading near-end information stored in a near-end setting component corresponding to the roll paper stored in the paper set component and storing the near-end information in the near-end setting component.

14. A thermal printer, comprising:

a thermal print head;

a storage component configured to store, for each of a plurality of types of roll paper, near-end information indicating whether to notify near-end of the roll paper;

a paper set component in which the roll paper is set;

a determination component configured to determine a type of the roll paper to be used;

a setting component configured to read near-end information relating to the roll paper, the type of which is determined, from the storage component and set the near-end information;

a near-end detection component configured to detect the near-end of the roll paper set in the paper set component; and

a near-end processing component configured to, only if the near-end information indicating that the near-end is notified is set by the setting component, according to detection of near-end of the roll paper by the near-end detection component, notify the near-end.

15. The thermal printer according to claim **14**, wherein

the plurality of types of roll paper comprise receipt roll paper obtained by winding long receipt paper, label roll paper obtained by aligning and winding a plurality of labels in a long direction on a long liner, and liner-less roll paper obtained by winding long liner-less paper,

the storage component stores, in association with the receipt roll paper, near-end information for notifying near-end and stores, in association with the label roll paper and the liner-less roll paper, near-end information for not notifying near-end, and

the setting component sets near-end information indicating that the near-end is notified if the receipt roll paper is determined by the determination component.

16. The thermal printer according to claim **14**, wherein, if near-end information indicating that the near-end is not notified is set by the setting component, the near-end processing component does not determine whether near-end of the roll paper is detected by the near-end detection component and does not notify the near-end.

17. The thermal printer according to claim **14**, wherein, if near-end information indicating that the near-end is not notified is set by the setting component, even if near-end of the roll paper is detected by the near-end detection component, the near-end processing unit does not notify the near-end.

18. The thermal printer according to claim **14**, further comprising a designation key that is operated in order to designate a type of roll paper, wherein

the determination component determines the roll paper for which the designation key operates.

19. The thermal printer according to claim **14**, wherein the near-end information comprises near-end disabling information and near-end enabling information.

20. The thermal printer according to claim **14**, wherein the determination component determines that when a receipt key is operated, the setting component reads near-end information stored in a near-end setting component corresponding to the roll paper stored in the paper set component and stores the near-end information in the near-end setting component.

* * * * *